

JUAN SEBASTIAN PEÑA DONNEYS

Madrid, Spain

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SUMMARY

Software engineer who builds and operates AI products end-to-end. Sole engineer on a production LLM agent that quotes and charges real customers, and technical founder of an anti-fraud ticketing platform.

EXPERIENCE

AI Software Engineer Intern Strategos AI

Madrid · Remote
Feb 2026 – Present

- Sole engineer on Craig, a production AI agent for a Dublin print shop that quotes, charges and manages customer orders end-to-end.
- Design and operate LLM multi-agent workflows against real business processes. Direct report to the CEO; the founders are non-technical.

Technical Founder Boletapp

Colombia
2026 – Present

- Anti-fraud ticketing platform (built, in testing, pre-launch): constantly rotating HMAC-derived QR codes and an offline-first door scanner that validates without connectivity.
- One cryptographic engine implemented twice, in TypeScript and Python, held bit-for-bit identical by shared test vectors.

GenAI Mavericks – hackathon (1 week) Accenture

Madrid
2025

- First edition, 100+ students from 20 universities: a RAG solution turning legal documents into plain language (Justice 2030 Spain initiative).

SELECTED PROJECTS

TikTok Recommendation Platform

Team of 7 · Jan–Apr 2026 · Python, FastAPI, React

- 98 of 282 commits, 22 pull requests opened and 9 of my teammates' PRs reviewed.
- Owned evaluation: NDCG@k ranking metrics, temporal splits respecting point-in-time boundaries, and 25 regression tests validating the metrics against analytically known ground truth.
- Built the video analysis pipeline (8 parallel branches: Whisper, OCR, captioning, keywords); cut it from over 5 minutes per video to under 60 seconds.

Semantic Engagement on TikTok – published in Towards AI

2026 · Python, transformers, FAISS

- Do semantically similar captions perform similarly? Across 22,647 posts, 12 Mann-Whitney U tests show 10.5–13% lower engagement variance than random baselines.

QR Forge

Open source · 2026 · FastAPI, React, Docker, Azure

- 136 tests at 80% line coverage behind a CI gate; three-stage non-root Docker image; CI/CD to Azure with Trivy, Bandit and Safety scanning.

EDUCATION

IE University BSc in Computer Science and Artificial Intelligence

Madrid, Spain
Expected: Jul 2027

- CUBICO Young Talented Leaders Scholarship (academic and leadership excellence). Machine Learning: 9.5/10.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C

AI & ML: LLMs and multi-agent systems, RAG, transformers and embeddings, FAISS, PyTorch, scikit-learn, computer vision

Backend & data: FastAPI, PostgreSQL, SQLAlchemy, Alembic, pandas

Infrastructure & frontend: Docker, GitHub Actions, Azure, GCP, Prometheus, React, Next.js, Tailwind

LANGUAGES

Spanish (native) · **English** (bilingual, daily professional use)